

```
---  
title: "Assignment 1 Code"  
output: html_document  
---
```

Overview

In this exercise I took a database with close to 500 observations for 25 variables. From this database I selected 50 observations for 5 variables. This was done basically in two steps. I eliminated the variables I am not interested in and selected only countries in Latin America.

Introduction

My ultimate aim is to analyze disparities across four child-related indicators in Latin America. The axes of disparities are: sex, urban-rural location, mothers' education, and sub-national political administrative units (admin level 1).

As a first step, in this exercise, I am putting together the data for birth registration. I am only including sex (male/female) disaggregation.

#Steps

In order to set the data up, I exported the data from the UNICEF Data Warehouse and saved them as a CSV file in Github.

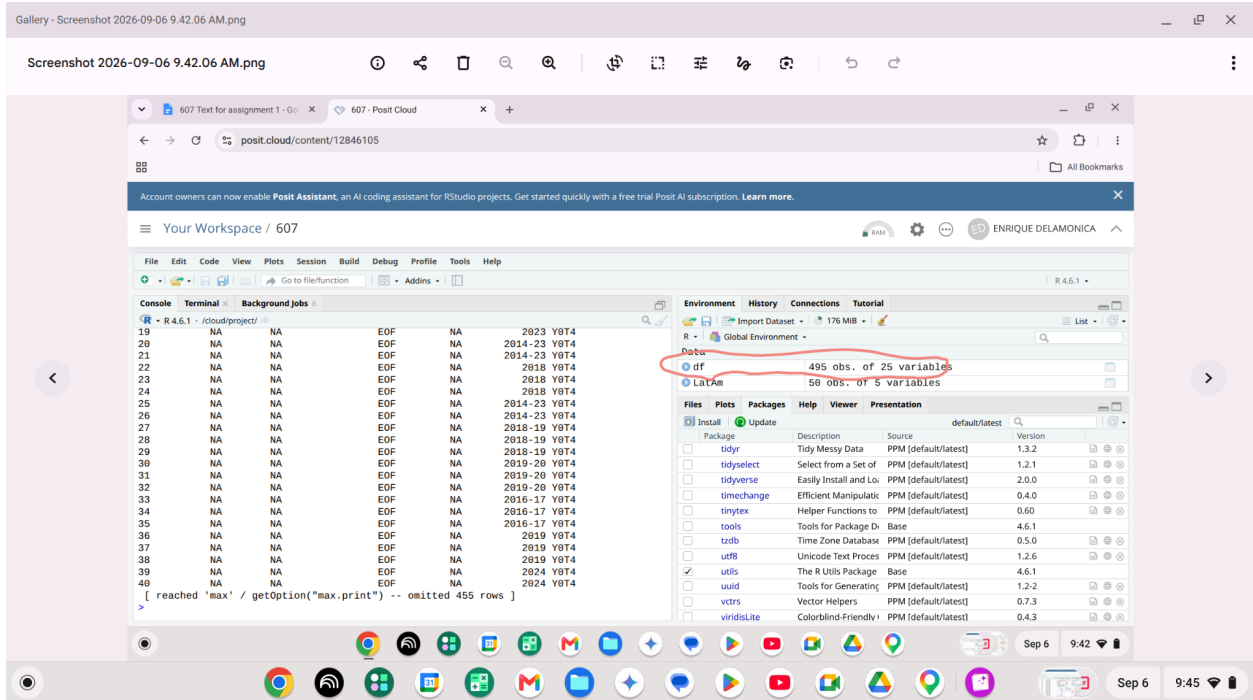
The url is:

```
"https://raw.githubusercontent.com/Enrique01234/607-Fall-2026/refs/heads/main/Copy%20of%20fusion_GLOBAL_DATAFLOW_UNICEF_1.0_.PT_CHLD_Y0T4_REG..csv"
```

Then, I brought the data from GitHub to RStudio. I followed the instructions in the assignment and using the url where I placed the data in GitHub:

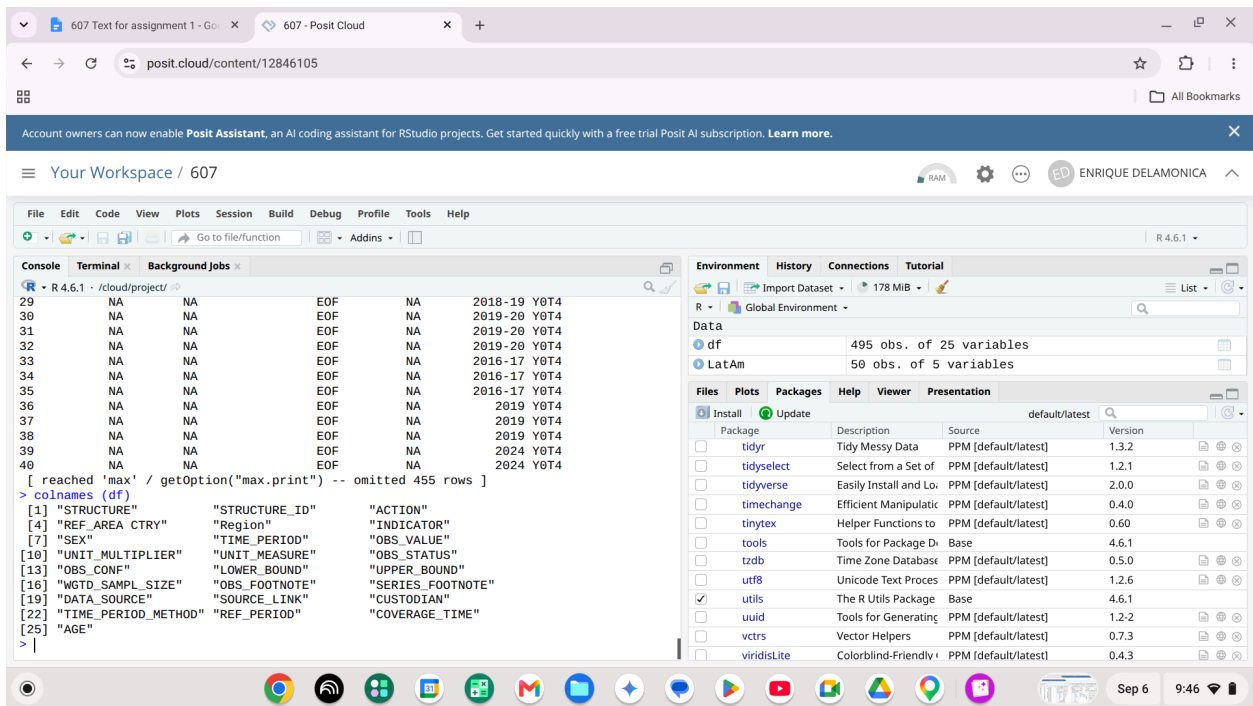
```
> url<-  
"https://raw.githubusercontent.com/Enrique01234/607-Fall-2026/refs/heads/main/Copy%20of%20fusion_GLOBAL_DATAFLOW_UNICEF_1.0_.PT_CHLD_Y0T4_REG..csv"  
> df <- read_csv(  
+   file = url,  
+   show_col_types = FALSE,  
+   progress = FALSE)
```

In the Environment Pane, I can see there are 495 observations and 25 variables in the data frame called "df".



In order to check the names of the columns (variables), I use

> colnames(df)



Now, not only can I see the names, I can also know their location: the columns are numbered according to their position in the data frame. This is what is needed to select the subset of variables I am interested in.

The columns I am interested in are: “REF_AREA CTRY” (which has the name of the country of the observation), “Region” (which has a numeric code for geographic regions such as South Asia or Western and Central Africa; Latin America is coded as 1), “TIME_PERIOD” (the year of the observation), “SEX” (a code shows if the value is for the whole country, “_T”, only boys, “M”, or only girls “F”), and “OBS_VALUE” (which is the actual number, the percentage of children whose birth is registered before they reach age 5).

I did this in two different ways

I specifically selected the columns I am interested in and created a new data frame called “new_dftotry”:

```
>new_dftotry <- df[,c(4, 5, 7, 8, 9)]
```

In the Environment Pane, I can see there are 495 observations and 25 variables in the data frame called “new_dftotry”.

I can see the result using

```
> new_dftotry
```

The screenshot shows the RStudio interface. The console on the left displays the following output:

```
R 4.6.1 - /cloud/project/ >
179      BLZ      1 _T      2016  95.700
180      BLZ      1 M      2016  95.300
181      MEX      1 M      2023  93.800
182      MEX      1 _T      2023  93.900
183      MEX      1 F      2023  93.900
184      BEN      2 F      2022  92.300
185      BEN      2 M      2022  92.800
186      BEN      2 _T      2022  92.600
187      UNICEF_MENA 0 _T      2023  89.330
188      BOL      1 F      2023  95.300
189      BOL      1 _T      2023  95.600
190      BOL      1 M      2023  95.800
191      MCO      4 F      2020 100.000
192      MCO      4 M      2020 100.000
193      MCO      4 _T      2020 100.000
194      MNG      6 F      2018  99.600
195      MNG      6 _T      2018  99.600
196      MNG      6 M      2018  99.600
197      BWA      3 F      2021 100.000
198      BWA      3 _T      2021 100.000
199      BWA      3 M      2021 100.000
200      MSR      3 _T      2017 100.000
[ reached 'max' / getOption("max.print") -- omitted 295 rows ]
>
```

The Environment pane on the right shows the following data frames:

- Global Environment
 - LatAm: 50 obs. of 5 variables
 - new_df: 495 obs. of 23 variables
 - new_df2: 495 obs. of 5 variables

The Packages pane shows the following installed packages:

Package	Description	Source	Version
tidyr	Tidy Messy Data	PPM [default/latest]	1.3.2
tidyselect	Select from a Set of	PPM [default/latest]	1.2.1
tidyverse	Easily Install and Lo	PPM [default/latest]	2.0.0
timechange	Efficient Manipulati	PPM [default/latest]	0.4.0
tinytex	Helper Functions to	PPM [default/latest]	0.60
tools	Tools for Package Di	Base	4.6.1
tzdb	Time Zone Database	PPM [default/latest]	0.5.0
utf8	Unicode Text Proces	PPM [default/latest]	1.2.6
utils	The R Utils Package	Base	4.6.1
uuid	Tools for Generating	PPM [default/latest]	1.2-2
vcdrs	Vector Helpers	PPM [default/latest]	0.7.3
viridisLite	Colorblind-Friendly	PPM [default/latest]	0.4.3

I also tried a longer way, by eliminating all variables except “REF_AREA CTRY”, “Region”, “TIME_PERIOD”, “SEX”, and “OBS_VALUE”. It might be easier to select the ones I am interested in, but I was exploring the select function and I wanted to check using the minus sign.

```
> new_df2 <- df[, -c(1, 2, 3, 6, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25)]
```

The new data frame, called "new_df2", has 5 columns and 495 observations, which I can check with:

```
> new_df2
```

It gives the same result as new_dftotry

The screenshot shows the RStudio interface. The console pane on the left displays the output of the R code, showing a data frame with 5 columns and 495 observations. The environment pane on the right shows the global environment with variables: LatAm (50 obs. of 5 variables), new_df (495 obs. of 23 variables), and new_df2 (495 obs. of 5 variables). The packages pane at the bottom right shows a list of installed and available packages, including tidyverse, tidyr, tidymodels, and others.

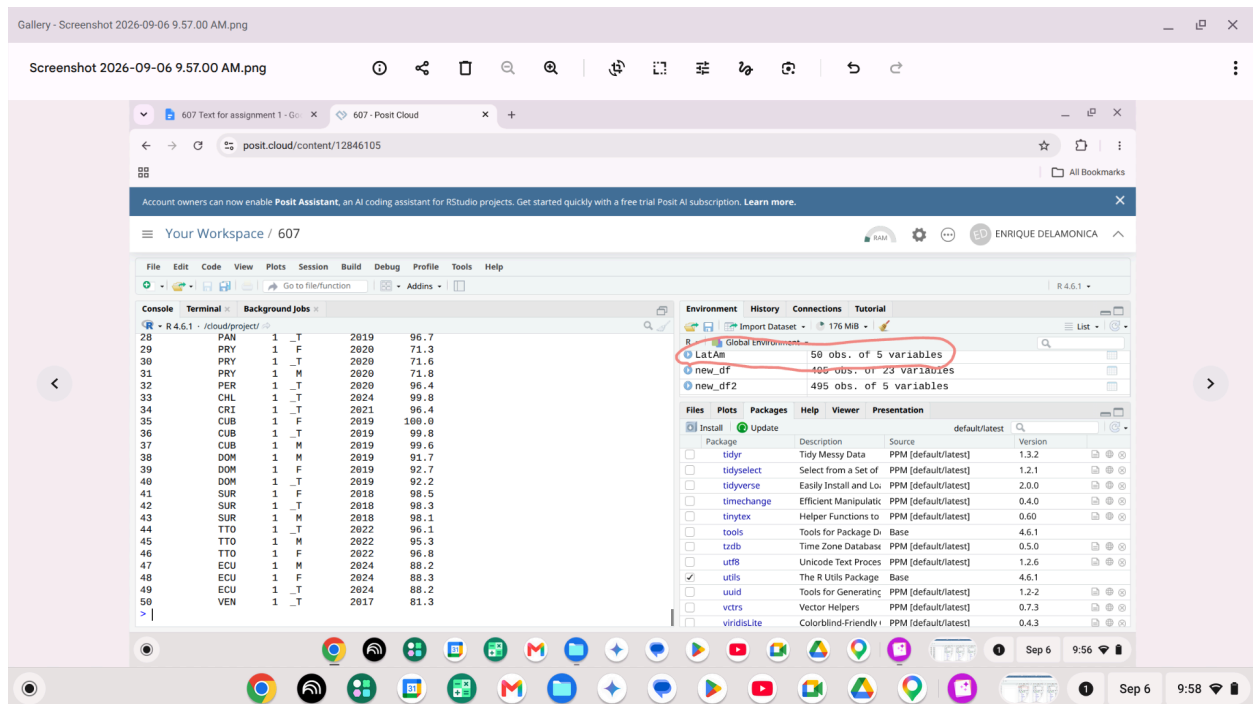
Please notice there is also a data frame called “new_df”. I created it while I was exploring the select function but it did not work as expected, I had not managed to delete all the variables I wanted to delete.

Finally, I selected only the countries in Latin America. The result is a new data frame called "LatAm".

```
> LatAm <- subset(new_df2,
+ Region == "1")
```

The new data frame has 5 columns and 50 observations, as can be seen in the Environment pane and using

> LatAm



#Conclusion

There are 50 observations in the LatAm data frame. Some of them are the only observation for a country. This is the case for Chile (CHL), Costa Rica (CRI), and Venezuela (VEN) as they appear only once and the SEX column shows “_T”. However, for most countries there are three observations (with information for the whole country, “_T” and disaggregated between boys, “M”, and girls, “F”). Using code, I would like to count the number of countries.

Countries without male/female disaggregation would have to be separated from the ones without disaggregation. However, they should not be deleted as the same countries may have disaggregation by urban-rural or other axes of disparity, which could be used for further analysis.

I would also like to check the most recent year with data. It seems to be 2023, but I want to find this out using code. Also, how many countries? How many countries with data in either 2023 or 2022?